

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 273

THE TELESURGERY SUITE

the two-pound surgeon, driven from anywhere
Lindbergh 2001 — spaceMIRA 2024



a third of US counties have no local surgeon

MINIRAS · THE WATCHES · THE PODS · THE ENVELOPE
SURGICAL REACH WITHOUT SURGICAL PRESENCE

Medical systems — v1.0 in force

GP-IM-273

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

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THE GEMINI PROTOCOLS — CHAPTER 273

PHASE 273: THE TELESURGERY SUITE — SHIP AND COUNTY — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING ([MEDICAL] TAG; THE PRECEDENTS ARE PUBLISHED: LINDBERGH 2001, SPACEMIRA 2024). The two-pound surgeon, driven from anywhere. The precedents are published history: the Lindbergh Operation (2001) removed a gallbladder across the Atlantic by robot; spaceMIRA (February 2024) flew the two-pound surgical robot to the space station. The suite assembles the file's own machines around that line: miniRAS at the table, Phase 166 vitals watches on the patient, Phase 163 triage pods as the room, Phase 246's envelope as the record. The ship fit: surgery at crossing distance, where the surgeon is a signal away and the pod is stocked. The Earth fit is the market and the mission: a third of US counties have no local surgeon — the same suite, driven from a city, closes the county gap. [MEDICAL] tag carried: the suite is medicine, graded as medicine, and the file's medical doctrine (the envelope's honesty, the band's dose-ledger) governs it.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-273, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the suite law as seated (contents line, the suite in crayon — the precedents, the ship fit, the Earth fit, the assembly — entire) — §2 the physics, graded — §3 the system in the suite — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 273: **The Telesurgery Suite — Ship and County.** miniRAS-class two-pound surgical robot driven from anywhere; Lindbergh 2001 to spaceMIRA 2024; integrates Phase 166 watches, Phase 163 triage pods, Phase 246 prill ice; serves the surgeon-less counties on Earth. [MEDICAL]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: surgical reach without surgical presence — a miniRAS-class robot on the ship (or in an underserved county) driven by a surgeon anywhere, integrated with the crew's seated medical stack.'

- **OLD, AND PROVEN (sitting 117):** the Lindbergh Operation, 2001 — a surgeon in New York removed a gallbladder in Strasbourg by robot, transatlantic.
- **CURRENT, AND SMALL:** February 2024 — spaceMIRA, a two-pound surgical robot a thousand times lighter than the industry tower, ran on the ISS with surgeons in Lincoln, Nebraska driving through the comm lag; the terrestrial MIRA won FDA clearance for colectomies the same month.
- **SHIP FIT (immediate):** miniRAS-class robot + Phase 166 vitals watches + Phase 163 triage pods + Phase 246 prill ice — the surgical suite assembles from seated phases. Crew fit: 4 surgeons and 2 dental surgeons drive it; the 4 medical assistants/nurses run the bay.
- **EARTH FIT:** a third of US counties have no local surgeon — the same box serves them; a Commercial Earth sibling under the open-hardware decree.

ORIGIN OF THOUGHT — PROVENANCE (VERBATIM)

Pre-approved by the author in sitting 117 (9 August 2026), number and classification assigned on his order of 11 August 2026: "telesurgery next number and to medical" — PHASE 273, MEDICAL.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PRECEDENTS ARE PUBLISHED, NOT PROMISED: Lindbergh 2001 — a gallbladder removed across the Atlantic by robot; spaceMIRA February 2024 — the two-pound surgical robot flown to the station. Telesurgery is not the speculative part; the suite is the file's assembly of proven parts.

THE SUITE IS AN ASSEMBLY OF THE FILE'S OWN MACHINES: miniRAS at the table, 166's vitals watches on the patient, 163's triage pods as the room, 246's envelope as the record. The phase invents the integration, not the instruments.

THE SHIP FIT: surgery at crossing distance — the surgeon a signal away, the pod stocked, the watches logging. Latency is priced honestly: the ship's crossing keeps the loop inside what the precedents prove.

THE EARTH FIT IS THE MARKET AND THE MISSION: a third of US counties have no local surgeon — the same suite, driven from a city, closes the county gap. The ship machine and the county machine are one build, and the Earth ledger pays for both.

THE [MEDICAL] TAG GOVERNS: the suite is medicine — graded as medicine, regulated as medicine, and held to the envelope's honesty: what it cannot do is stated on the same page as what it can.

§3 — THE SYSTEM IN THE SUITE

- **The table:** miniRAS — the two-pound surgeon, driven from anywhere.
- **The patient:** Phase 166 vitals watches — the monitor already worn.
- **The room:** Phase 163 triage pods — the sterile cell already drawn.
- **The record:** Phase 246's envelope — the dose-ledger honesty as the chart.
- **The precedents:** Lindbergh 2001, spaceMIRA 2024 — published history, not promise.
- **The fits:** ship at crossing distance; Earth where a third of counties have no surgeon.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phases 163, 166 and 246 are the family the suite assembles: the triage pods, the vitals watches, the envelope — the file's medical stack, integrated at the table.
- The precedents are the file's citation habit: Lindbergh and spaceMIRA named with dates — the claim stands on published history.
- The Earth fit follows the file's dual-market doctrine: the ship machine priced so the county machine pays for it — the settlement economics running through medicine.
- The [MEDICAL] tag ties the suite to the file's medical law: the envelope's honesty and the band's ledger govern every line.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Bench the table: miniRAS driven over the ship's own loop — latency measured, not assumed.
- 2. Wire the monitors: 166's watches to the suite's chart — the patient's numbers in the surgeon's hands.
- 3. Fit the room: a 163 triage pod stocked to the surgical list — the sterile cell proven before the patient.
- 4. Run the envelope: 246's record format as the chart — what happened, logged; what is owed, named.
- 5. Price the county unit: the Earth fit costed against the county gap — the ledger that pays for both.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 163 (the triage pods):** the room — the sterile cell the suite operates in.
- **Phase 166 (the vitals watches):** the monitor — the patient's numbers already on the wrist.
- **Phase 246 (the envelope):** the record — the chart format and the honesty law.
- **Phases 264/269 (the bank):** the supply — the suite's draw priced against the station power.

§8 — DO-NOT-BUILD

- Do NOT oversell the latency — the loop is priced against the precedents; beyond it, the suite says no.
- Do NOT improvise the room — a stocked 163 pod or nothing; surgery is not a corridor.
- Do NOT drop the tag — [MEDICAL] governs: graded as medicine, regulated as medicine.
- Do NOT forget the chart — 246's envelope is the record; an unlogged cut never happened.
- Do NOT price the ship without the county — the Earth fit pays for both; the doctrine is the dual ledger.

§9 — OWED

OWED: the bench list, whole — the latency bench over the ship loop; the surgical stocking list for the pod; the watch-to-chart integration; the county unit's pricing; the regulatory path named. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-273, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING ([MEDICAL] TAG; THE PRECEDENTS ARE PUBLISHED: LINDBERGH 2001, SPACEMIRA 2024), seated law, master v2.1 (numerical print order). Built 24 August 2026, fourteenth of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20

while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 273